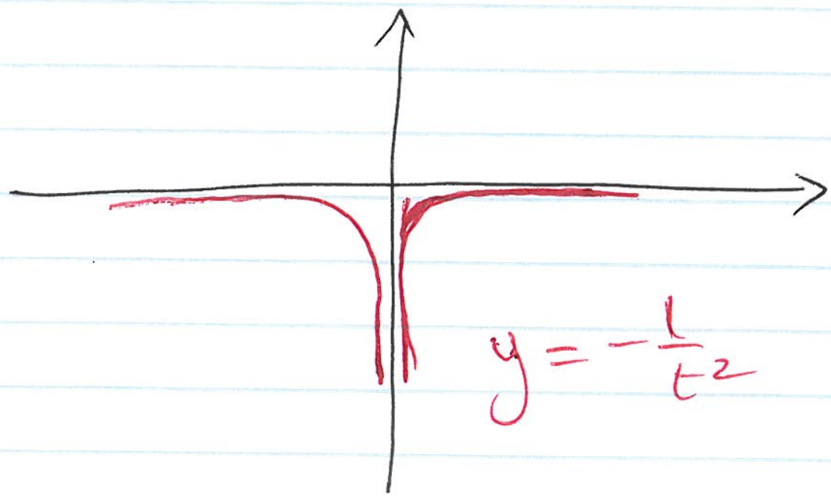
(c) $f(t) = -\frac{1}{t^2}$

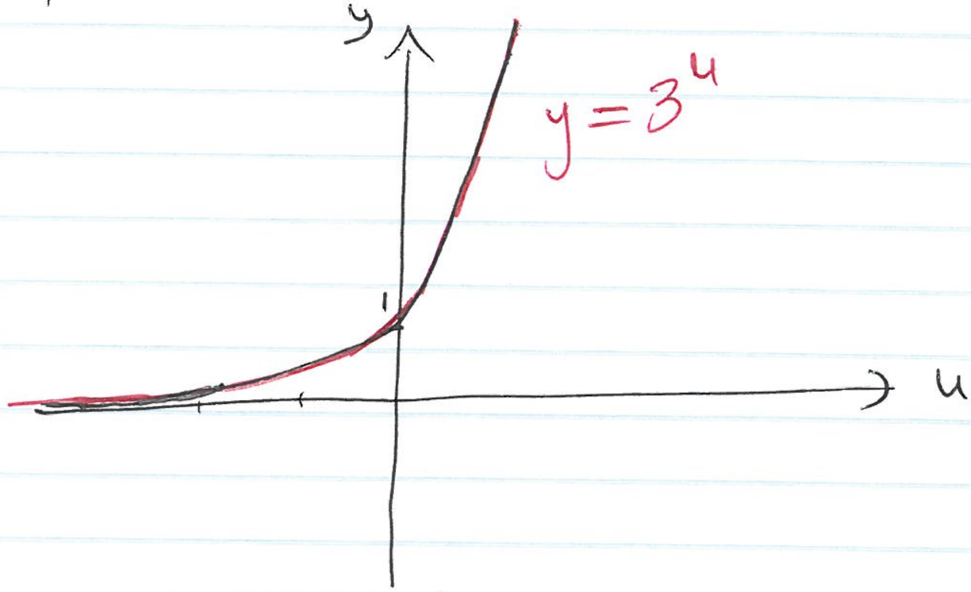
$D_f = \{ t \in \mathbb{R} \ / \ t \neq 0 \} = (-\infty, 0) \cup (0, +\infty)$



$R_f = (-\infty, 0)$

(d) $f(u) = 3^u$

$D_f = \mathbb{R} = (-\infty, \infty)$



$R_f = (0, +\infty)$

Example 5: Determine the domain of

$$f(r) = \frac{1}{2r^2 - 3} \quad ; \quad g(\sigma) = 3\sigma^2 - 2\sigma$$

Definition: A function defined by different formulas in different parts of ~~itest~~ its domain is called piecewise defined functions

Example 6:

(a)
$$f(x) = \begin{cases} 4 , & \text{if } x < -2 \\ -x^2 + 1 , & \text{if } -2 \leq x < 1 \\ x + 4 & \text{if } x \geq 1 \end{cases}$$

(b)
$$\sqrt{|x|} = \begin{cases} \sqrt{x} , & \text{if } x \geq 0 \\ \sqrt{-x} & \text{if } x < 0 \end{cases}$$

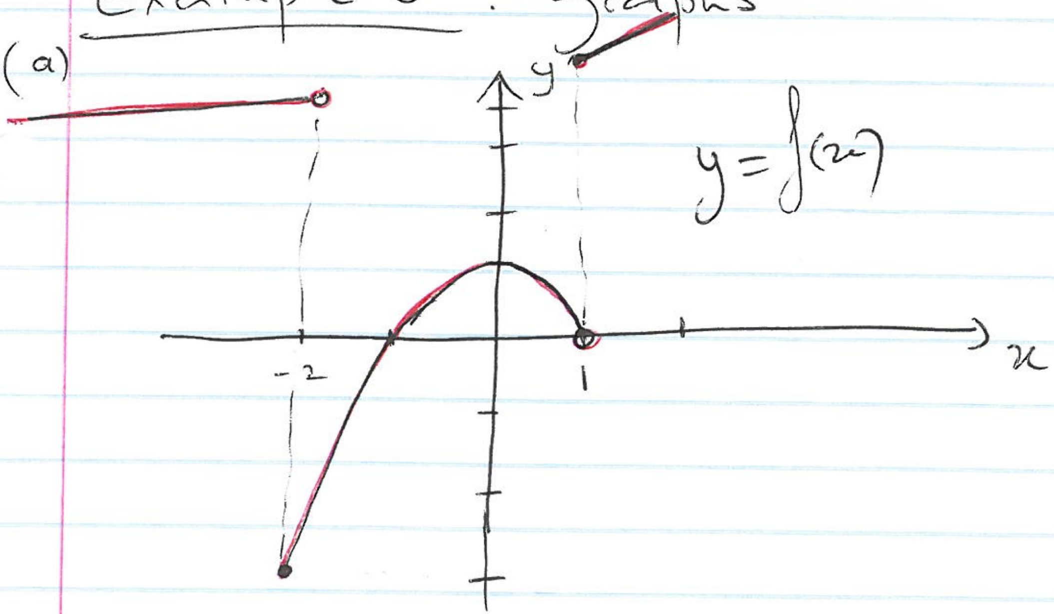
(c) $h(x) = x + |x+1|$; $h(x) = \begin{cases} 2x+1 , & x \geq -1 \\ -1 , & x < -1 \end{cases}$

(d)
$$g(x) = \begin{cases} \frac{1}{x^2 - 1} , & \text{if } x \geq 2 \\ 2x + 3 , & \text{if } x < 2 \end{cases}$$

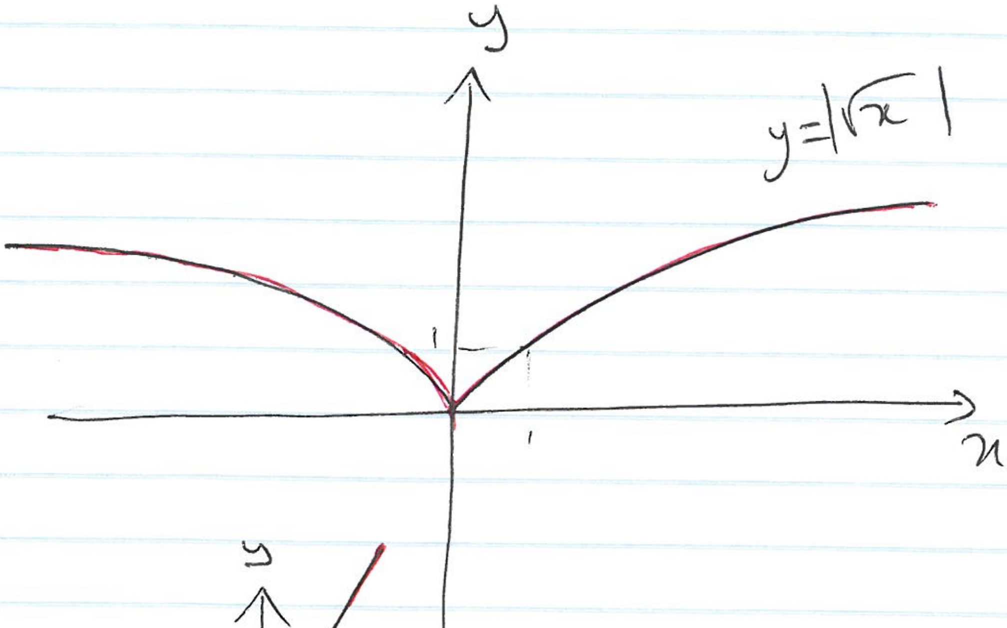
Example 7: Determine the domain of the function in Eg 6, sketch their graphs and find their respective ranges.

Example 6 : graphs

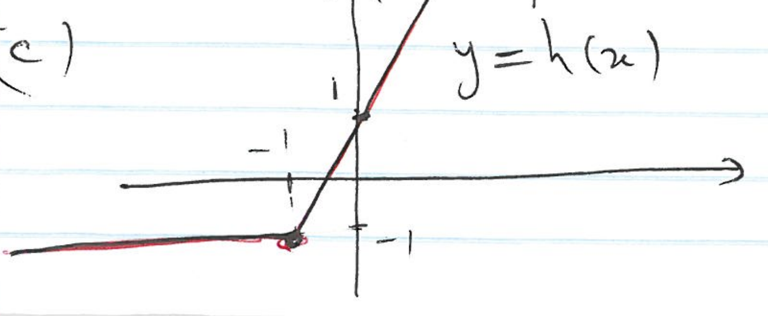
(a)



(b)



(c)



Example 5 :

$$D_f = \{ r \in \mathbb{R} \;/\; 2r^2 - 3 \neq 0 \}$$

$$2r^2 - 3 = 0 \iff r^2 = \frac{3}{2} \iff r = \pm\sqrt{\frac{3}{2}}$$

so

$$D_f = \left(-\infty, -\sqrt{\tfrac{3}{2}}\right) \cup \left(-\sqrt{\tfrac{3}{2}}, \sqrt{\tfrac{3}{2}}\right) \cup \left(\sqrt{\tfrac{3}{2}}, +\infty\right)$$

$$D_g = (-\infty, +\infty)$$

Example 7 :

(a) $D_f = \mathbb{R} = (-\infty, +\infty)$ since all the pieces are well defined on each of their corresponding intervals

(b) Observe that since $|x| \geq 0$ then $\sqrt{|x|}$ is defined for all $x \in \mathbb{R}$

$$D = (-\infty, +\infty)$$

(c) $D_h = (-\infty, +\infty)$

(d) $D_g = (-\infty, +\infty)$

Example 8 : sketch the graph of $y = |f(x)|$

(a) $f(x) = \sin x$, $x \in [-\pi, \pi]$

(b) $f(x) = 2x - 1$

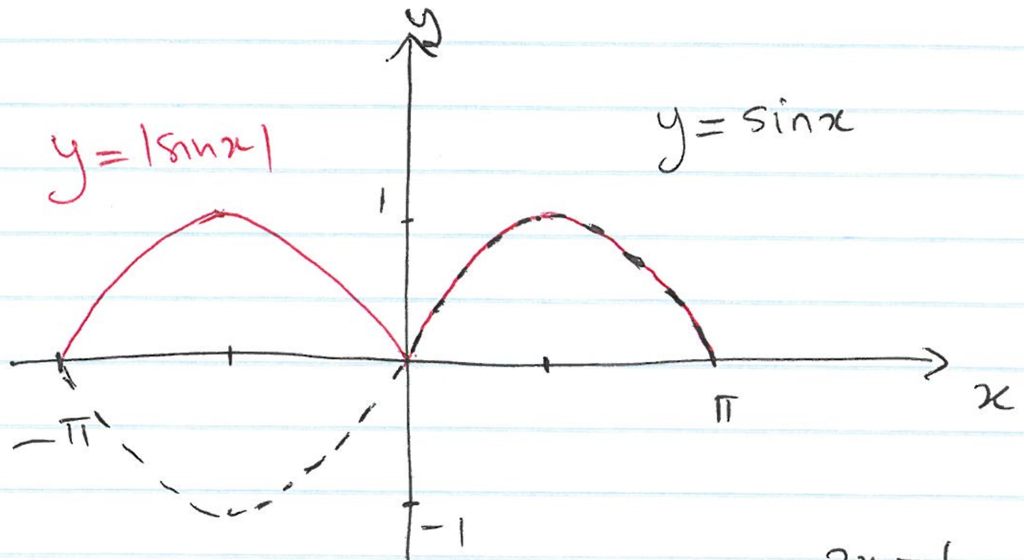
(c) $f(x) = \frac{1}{x}$

(d) $f(x) = -\frac{1}{x^2}$

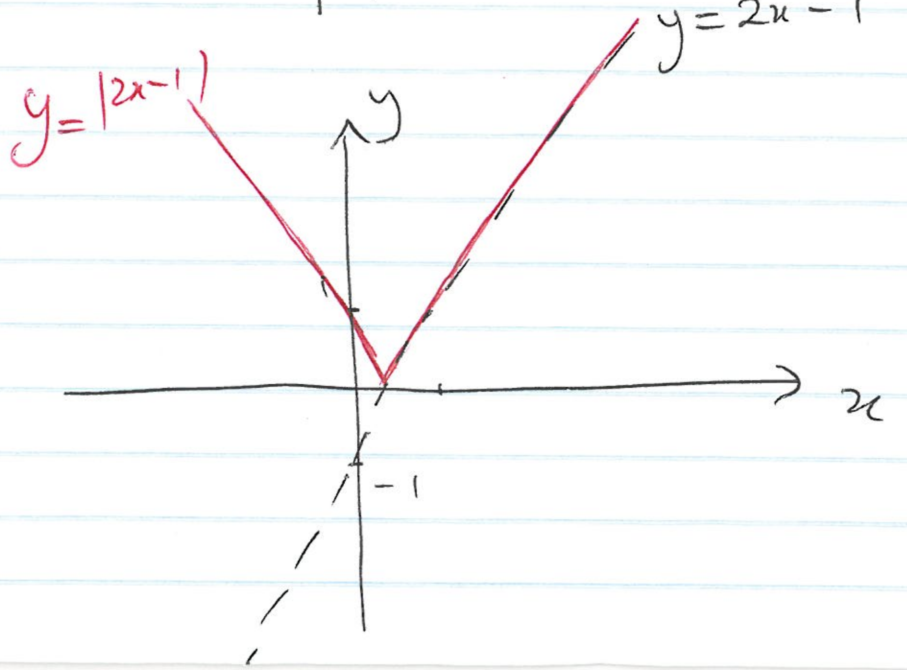
(e) $f(x) = 3x^2 - 2x$

(f) $f_1(x) = 3^x$, $f_2(x) = \sqrt{-x}$, $f_3(x) = \sqrt{x}$

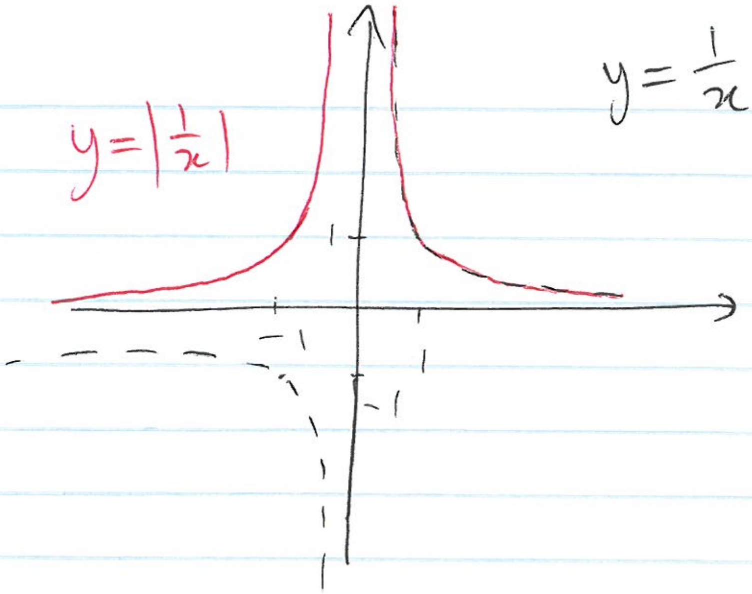
$\therefore$ (a)



(b)

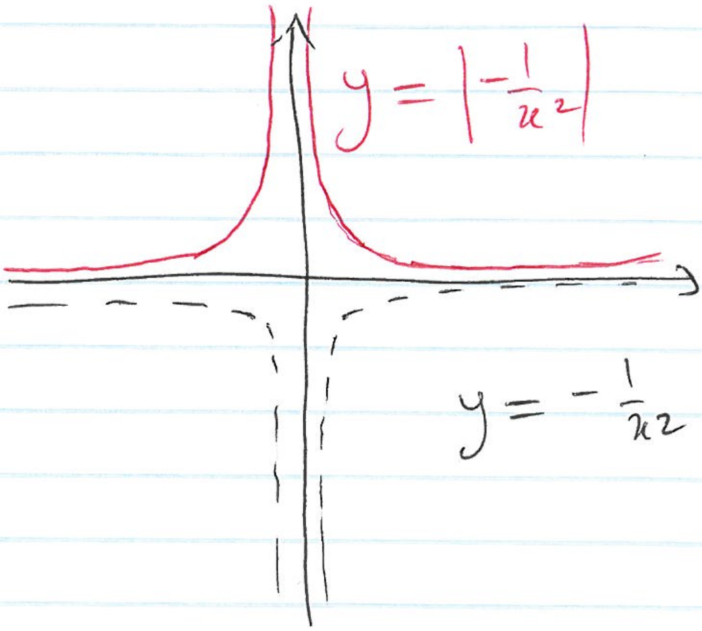


(c)



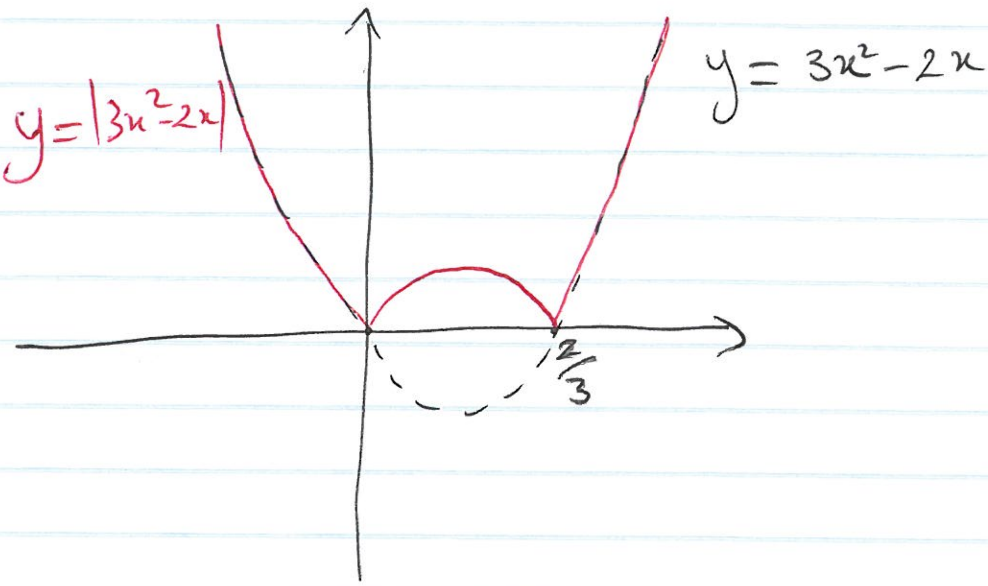
$y = \frac{1}{x}$
$y = \left|\frac{1}{x}\right|$
1
-1
-1
1

(d)



$y = \left|-\frac{1}{x^2}\right|$
$y = -\frac{1}{x^2}$

(e)



$y = |3x^2 - 2x|$
$y = 3x^2 - 2x$
$2/3$

(f) Observe that all these functions are greater or equal to zero so the graph of $y_1 = f(x)$ and $y_2 = |f(x)|$ are indentical.

Definition :

A function f is called even if and only if for all $x \in Df$ then $-x \in Df$ and

$$f(-x) = f(x)$$

A function f is called odd if and only if for all $x \in Df$ then $-x \in Df$ and

$$f(-x) = -f(x).$$

Graphically, the graph of an even function is symmetric with respect to the y-axis, whereas the graph of an odd function is symmetric about the origin.

Example 9 :

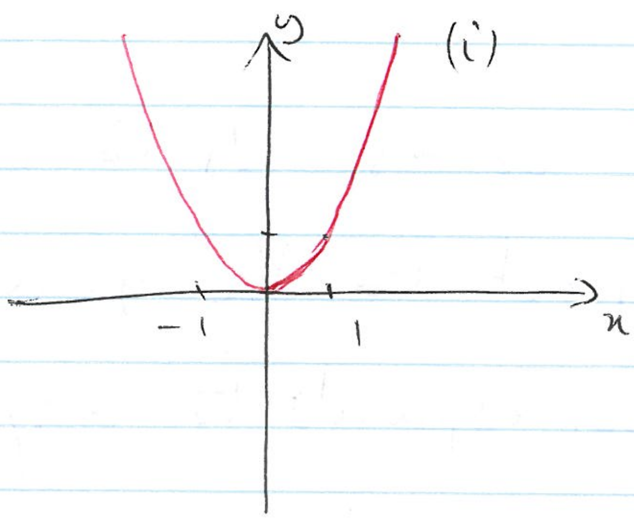
(a) (i) $y = x^n$, with n even, $n \in \mathbb{N}$

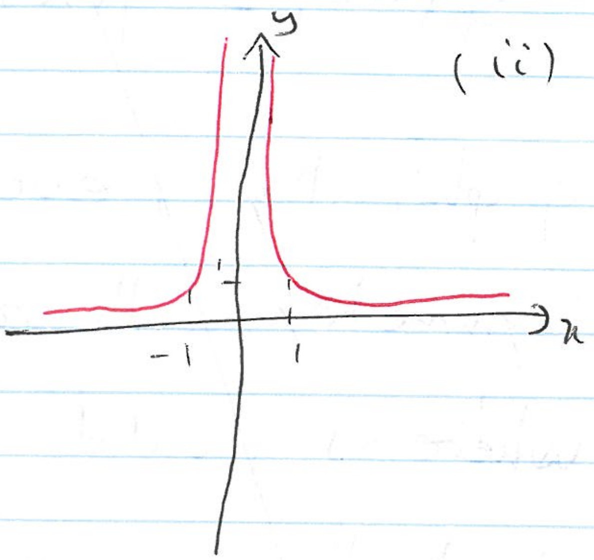
(ii) $y = x^{-n}$, with n even, $n \in \mathbb{N}$

(b) (i) $y = x^n$, with n odd, $n \in \mathbb{N}$

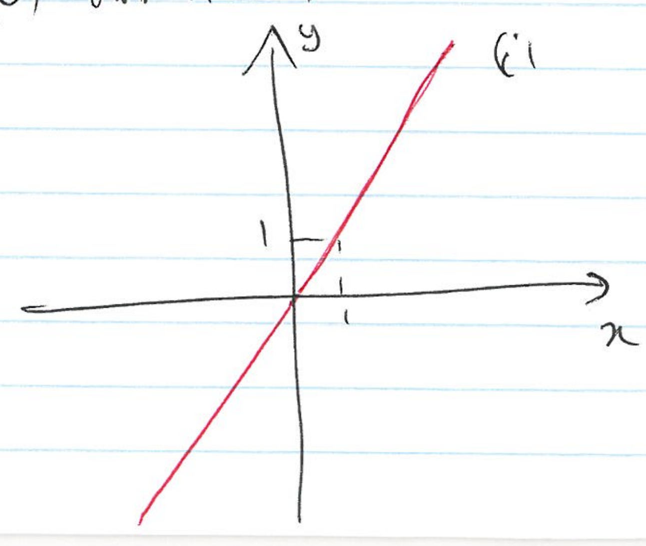
(ii) $y = x^{-n}$, with n odd, $n \in \mathbb{N}$

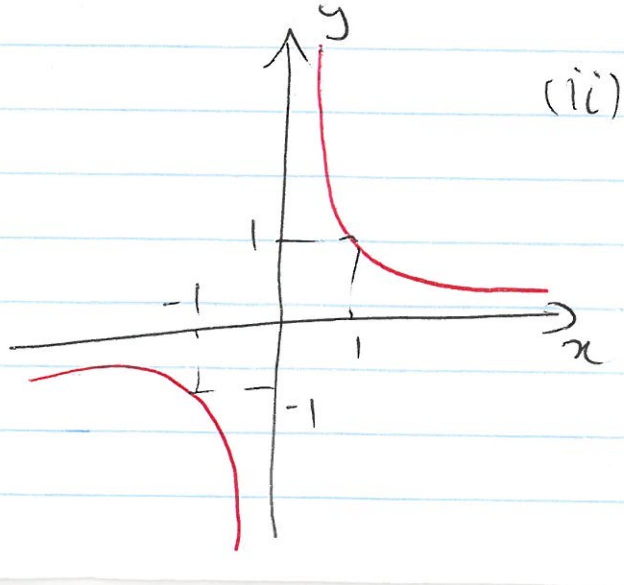
(a) with $n = 2$





(b) with $n = 1$





(a) let $n = 2m \quad , \quad m \in \mathbb{N}$

(i) $f(x) = x^{2m} \quad ; \quad f(-x) = (-x)^{2m} = (-1)^{2m} \cdot x^{2m} = x^{2m}$

$f(-x) = f(x) \quad \forall x \in \mathbb{R}$

(ii) $f(x) = \dfrac{1}{x^{2m}} \quad ; \quad f(-x) = \dfrac{1}{(-x)^{2m}} = \dfrac{1}{(-1)^{2m} \cdot x^{2m}} = \dfrac{1}{x^{2m}}$

$f(-x) = f(x)$

(b) let $n = 2m+1 \quad , \quad m \in \mathbb{N}$

(i) $f(x) = x^{2m+1} \quad ; \quad f(-x) = (-x)^{2m+1} = (-1)^{2m+1} \cdot x^{2m+1} = -x^{2m+1}$

$f(-x) = -f(x)$

(ii) $f(x) = \dfrac{1}{x^{2m+1}} \quad ; \quad f(-x) = \dfrac{1}{(-x)^{2m+1}} = -\dfrac{1}{x^{2m+1}}$

$f(-x) = -f(x)$

Most functions are neither even nor odd.

Example 10 : Are the followings functions even, odd or neither?

$f(x) = \dfrac{-x^2}{x^2+1}$; $g(t) = -t^2-4$; $g_2(x) = 5^x$

$h(x) = x^2+2x+1$; $f_2(x) = 5x^3+2x$

• $D_f = (-\infty, +\infty)$ (since $x^2+1 \neq 0$)

$f(-x) = \dfrac{-(-x)^2}{(-x)^2+1} = \dfrac{-x^2}{x^2+1} = f(x)$

$\Longrightarrow$ f is even

• $D_g = (-\infty, +\infty)$

$g(-t) = -(-t)^2-4 = -t^2-4 = g(t)$

• $D_h = (-\infty, +\infty)$

$h(-x) = (-x)^2+2(-x)+1 = x^2-2x+1 \neq h(x)$ or $h(-x)$

h is neither odd nor even

• $D_{f_2} = (-\infty, \infty)$

$f_2(-x) = 5(-x)^3+2(-x) = -5x^3-2x = -(5x^3+2x)$

$= -f_2(x)$

f_2 is odd

- $D_{g_2} = (-\infty, +\infty)$

$g_2(-x) = 5^{-x} = \frac{1}{5^x} \neq g_2(x)$ or $g_2(-x)$

g_2 is neither odd nor even.

Observe that $f(x) = \sqrt{x}$ is such that $D_f = [0, +\infty)$ so if $+x \in D_f$, $-x \notin D_f$. It is then useless to check if the function is even or odd.